S. Harinesh

192521449

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SIMATS ENGINEERING

ASSESSMENT TOOL 3

MCQ Test

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 20%

QUESTIONS:20 **Total Marks:20**

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1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs?
   a) Slow Start
   b) Fast Retransmit
   c) Fast Recovery
   d) Tahoe Reset
   (BL4)(1)
2. A TCP sender has CWND = 8 MSS and SSTHRESH = 16. After one RTT with no loss, CWND becomes:
   a) 9 MSS
   b) 16 MSS
   c) 12 MSS
   d) 8 MSS
   (BL4)(1)
3. Which field in the TCP header is used to implement flow control at the receiver side?
   a) Sequence number
   b) Acknowledgement number
   c) Window size
   d) Urgent pointer
   (BL4)(1)
4. Nagle's algorithm causes increased latency primarily because it:
   a) Disables ACK piggybacking
   b) Buffers small segments until ACK arrives
   c) Reduces CWND on every packet

d) Increases SYN backlog
(BL4)(1)

5. In a high-latency satellite link (600 ms RTT), TCP throughput is prima
limited by:
a) MTU size
b) CWND growth rate relative to RTT
c) DNS resolution time
d) UDP competition
(BL4)(1)

6. UDP is preferred over TCP for DNS queries primarily because:
a) UDP supports fragmentation
b) DNS responses are small and connection overhead is unnecessary
c) TCP cannot use port 53
d) UDP has better error correction
(BL4)(1)

7. A UDP-based streaming application experiences 15% packet loss. The bes
application-layer mitigation is:
a) Increase UDP buffer size
b) Switch to TCP
c) Implement Forward Error Correction (FEC)
d) Reduce frame rate to 1 fps
(BL4)(1)

8. What is the maximum UDP payload size without IP-level fragmentation on a standard Ethernet link (MTU = 1500 bytes)?
a) 1480 bytes
b) 1472 bytes
c) 1460 bytes
d) 1500 bytes
(BL4)(1)

9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is:
a) Eliminating IP header overhead
b) 0-RTT or 1-RTT connection setup
c) Removing encryption
d) Native multicast support
(BL4)(1)

10. HTTP/1.1 persistent connections reduce latency by:
a) Enabling UDP fallback
b) Reusing the same TCP connection for multiple requests
c) Compressing HTTP headers
d) Sending requests without waiting for DNS
(BL4)(1)

11. Head-of-line blocking in HTTP/1.1 occurs when:
a) The DNS server is slow

b) A response blocks subsequent responses on the same connection
c) TCP CWND drops to zero
d) The server rejects pipelined requests
(BL4)(1)

12. HTTP/2 multiplexing solves HOL blocking at the HTTP layer but not at the:
a) Application layer
b) DNS layer
c) TCP layer
d) IP layer
(BL4)(1)

13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks?
a) 301
b) 302
c) 304
d) 307
(BL4)(1)

14. In HTTP caching, the Cache-Control: no-cache directive means:
a) Never cache this resource
b) Validate with the server before using a cached copy
c) Cache for 0 seconds
d) Disable all caching headers
(BL4)(1)

15. Which DNS record type maps a domain name to an IPv6 address?
a) A
b) CNAME
c) AAAA
d) MX
(BL4)(1)

16. A DNS resolver switches from UDP to TCP when:
a) The query type is MX
b) The response is truncated (TC bit set)
c) TTL exceeds 3600 s
d) The authoritative server is unreachable
(BL4)(1)

17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID?
a) ARP poisoning
b) DNS cache poisoning
c) SYN flooding
d) BGP hijacking
(BL4)(1)

18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases:
a) Cache hit ratio

b) Recursive resolver query load
c) DNSSEC signature validity
d) Zone transfer frequency
(BL4)(1)

19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is:
a) HTTP → TCP → DNS
b) DNS → TCP handshake → HTTP GET
c) TCP → DNS → HTTP
d) DNS → HTTP → TCP
(BL4)(1)

20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game?
a) TCP for game state, HTTP for assets
b) UDP with application-layer sequencing for game state, HTTP/3 for assets
c) HTTP/2 for all traffic
d) TCP with Nagle enabled for all traffic
(BL4)(1)

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions